BASS STABLE RANK OF A DIRICHLET-SERIES ALGEBRA

ØISTEIN SØVIK

AI assistance was used in adapting and checking the argument and the exposition.

Sasane's Conjecture 6.1 asks: "Does $R(S) = \infty$ imply that the Bass stable rank of $\mathscr{H}_S^\infty$ is infinite?" Here $S \subseteq \mathbb{N}$ is a multiplicative subsemigroup containing 1, $R(S)$ is the supremum of the sizes of independent subsets, and independence means that no member lies in the subsemigroup generated by the others.

Here $S \subseteq \mathbb{N}$ is a multiplicative semigroup containing 1, $\mathscr{H}_S^\infty$ is the algebra of bounded Dirichlet series supported on S, and $R(S)$ is the supremum of the sizes of subsets in which no member belongs to the semigroup generated by the other members.

The classical theorem used here is Treil, *The stable rank of the algebra H^∞ equals 1*, Theorem 1, doi:10.1016/0022-1236(92)90015-B. Sasane, *On Banach subalgebras of the Dirichlet Hardy algebra $\mathscr{H}^\infty$*, doi:10.1007/s00020-026-02825-1.

Answered: no. Evidence: Sketch.

No. For $S_2 = \{2^k : k \geq 0\}$, $R(S_2) = \infty$ but $\operatorname{bsr}(\mathscr{H}_{S_2}^\infty) = 1$.

Proof. More generally, if $a > 1$ lies in S, then for each $M \geq 2$ the set $E_M = \{a^k : M \leq k < 2M\}$ is independent. A product of two or more of its elements has exponent at least $2M$, and a product of one other element cannot equal the omitted power. Thus independent sets have arbitrarily large finite size.

For S_2, the map $Ug(s) = g(2^{-s})$ is an isometric algebra isomorphism from $H^\infty(\mathbb{D})$ onto $\mathscr{H}_{S_2}^\infty$: the half-plane map $s \mapsto 2^{-s}$ has image $\mathbb{D} \setminus \{0\}$, and a bounded series supported on powers of two is a Taylor series in 2^{-s}. Treil's theorem gives $\operatorname{bsr}(H^\infty(\mathbb{D})) = 1$. Bass stable rank is invariant under unital algebra isomorphism, proving the counterexample. The source's additional unique-product generator condition is therefore essential to its positive theorem. □

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